Angela Zhao

(667) 464-5025
azhao29@jh.edu

Education

Johns Hopkins University, Bloomberg School of Public Health — Expected May 2024
ScM in Biostatistics — GPA: 3.93/4.0

McGill University — October 2022
BSc in Joint Honours Computer Science and Biology, Minor in Mathematics — cGPA: 3.97/4.0

Summary

4 years research experience	1 peer-reviewed article	5 leadership positions
4 years teaching experience	8 talks and posters	7 awards and honors

Publications

1. **Zhao A**, Cui E, Leroux A, Lindquist M, Crainiceanu C (2023). Evaluating the prediction performance of objective physical activity measures for incident Parkinson's disease in the UK Biobank. *Journal of Neurology,* 270(12), 5913–5923. https://doi.org/10.1007/s00415-023-11939-0
2. **Zhao A**, Cui E, Leroux A, Zhou X, Muschelli J, Lindquist M, Crainiceanu C. Objective physical activity is the strongest modifiable predictor of incident Alzheimer's disease in the UK Biobank *(submitted).*
3. Cui E*, **Zhao A***, Crainiceanu C, Leroux A. Functional and Distributional Cox Models for UK Biobank Physical Activity Analysis *(in preparation).*
4. **Zhao A**, Lindquist M, Caffo B. Comparing stochastic model performance with applications to fMRI data *(in preparation).*

Editorial Activities

Reviewer — *Journal of Neurology (1)*
Biostatistics and Epidemiology (1)

Research Experience

Johns Hopkins University — Baltimore, MD

Research Assistant, **Supervisor: Dr. Brian Caffo** — July 2023 - Present

Skills: Python (pytorch), R

- Evaluated and diagnosed stochastic models – including RNNs, transformers, and AR models – for regional fMRI data from the Human Connectome Project.

Research Assistant, **Supervisor: Dr. Ciprian Crainiceanu** (thesis advisor) — September 2022 - Present

Skills: R, JAGS, cluster computing

- Leveraged UK Biobank, a cohort study of 500,000+ participants, to study rare diseases and mortality risk.
- Produced generalizable software for functional and scalar Cox regression, sensitivity analysis, statistical inference, and model selection using UK Biobank data.
- Used MCMC to predict PD and mortality in the UK Biobank.

McGill University — Montréal, QC

Research Assistant, **Supervisor: Dr. Jackie Vogel** — January 2020 - August 2022

Skills and software: Java, Python, Dragonfly, ImageJ, Matlab

- Project 1: Developed a novel shape analysis pipeline for super-resolution microscopy data. Skills: signal processing of *in vivo* Kar9 protein trajectories; WEKA machine learning classification algorithms for artefact removal.
- Project 2: Single-cell analysis of spindle mechanics in different budding yeast mutants. Skills: weighted-mean and Gaussian sub-pixel positioning of image data, hypothesis testing.
- Wet lab: confocal and structured-illumination microscopy (Elyra7); preparing cell cultures.

Teaching Experience

Johns Hopkins University

Teaching Assistant, Statistical Programming Paradigms and Workflows, ∼45 students — 2023
Teaching Assistant, Statistical Methods in Public Health II, ∼300 students — 2023
Teaching Assistant, Statistical Machine Learning, ∼50 students — 2023

McGill University

Undergraduate Tutor, ∼10 students — 2020-2022
Subjects: Discrete Maths, Algorithms and Data Structures, Organic Chemistry I, and more
Teaching Assistant, Data Science, ∼200 students — 2021
Grader, Introduction to Biophysics, ∼150 students — 2020

Technical Skills and Coursework

Languages: R, Python, SAS, Java, C, Bash, MATLAB
Developer Tools and Software: ImageJ, Dragonfly, Git and GitHub, LaTeX, Linux Shell
Math and Computing: MCMC, Classification, regression, cluster computing
Cloud/Databases: REDCap, SQL
Highlighted Coursework
Undergraduate: Real Analysis I-II, Multivariate Calculus, Linear Algebra, ODE's, Algorithms and Data Structures, Artificial Intelligence, Computational Methods and Research in Biology, Data Science
Graduate: Essentials of Probability and Statistical Inference I-IV, Methods in Biostatistics I-IV, Statistical Machine Learning, Survival Analysis, Longitudinal and Multi-Level Models, Causal Inference, Statistical Computing

Posters and Abstracts

1. Morelli K, Meier SM, **Zhao A**, Willis M, Barral Y, Vogel J. Interrogating the material properties of a microtubule (+) end-associated nanoscale condensate in living cells using super-resolution microscopy, *American Society for Cell Biology Cell Bio Virtual*, 2023
2. Sim S, **Zhao A**, Vogel J. Phosphoregulation of y-tubulin regulates anti-parallel microtubule bundling through kinesin-5, *8th Meeting of the International Union of Microbeam Analysis Societies*, 2023
3. Sim S, **Zhao A**, Vogel J. y-Tubulin phosphoregulation controls interpolar microtubule organization through kinesin-5, *Mitotic spindle: From living and synthetic systems to theory*, 2023
4. Sim S, Morelli K, **Zhao A**, Vogel J. How does y-tubulin phosphoregulation control microtubule identity during spindle assembly?, *4th Annual CRBS Symposium*, Montreal, Canada, 2022
5. Sim S, Morelli K, **Zhao A**, Vogel J. Phosphoregulation of y-Tubulin Directs Spindle Assembly Through Kinesin-5 and Microtubule Number, *American Society for Cell Biology Cell Bio Virtual*, 2021

Presentations

1. Sim S, Morelli K, **Zhao A**, Vogel J. L'effet de la phosphorylation de la tubuline-y sur la kinésine-5 et sur la formation de paires de microtubules dans le fuseau mitotique, *90e Congres de l'ACFAS*, Université de Montréal, May 2023
2. **Zhao A**, Cui E, Leroux A, Lindquist M, Crainiceanu C. Physical activity to predict Parkinson's disease in the UK Biobank, *8th Annual Joint Session of the BHS Lab, ENGAGE Lab, and SMART-WIT groups*, Johns Hopkins Univeristy, April 2023
3. Sim S, Morelli K, **Zhao A**, Vogel J. Le rôle de la phosphorylation de la tubuline-y dans l'organisation des microtubules du fuseau mitotique médiée par la kinésine-5, *89e Congrès de l'ACFAS*, Université de Montréal, May 2022

Leadership Positions

Department of Biostatistics, Johns Hopkins University
ScM Student Representative 2023-2024
Curriculum Committee Representative, *Biostatistics Students Organization* 2023-2024
CEPH Site Visit Student Representative 2023
For reaccreditation from the Council on Education for Public Health
McGill Integrative Biosciences Society, McGill University
President 2021-2022
VP Events 2020-2021

Awards and Honors

Master's Tuition Scholarship, Johns Hopkins University 2023
Dean's Multidisciplinary Undegraduate Research List, McGill University 2022
Dean's Honour List, McGill University 2019-2022
Wing Hing Chan Scholarship in Science, McGill University 2021
Undergraduate Science Research Award, 2021
The Natural Sciences and Engineering Research Council of Canada (NSERC), $7500
Tomlinson Engagement Award for Mentoring, McGill University 2020-2021
J W McConnell Scholarship, McGill University 2018

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